

OPTIMUS PRIME / EXPEDITION SKILLS FOR HUMANOID MOUNTAINEERING

A SLIP BECOMES A RESCUE.

G1 MUST RECOVER BEFORE A HUMAN MUST INTERVENE.

Extreme terrain turns small failures into high-consequence missions.



FOUR SKILLS FOR EXTREME TERRAIN.

ONE EXPEDITION SYSTEM / CURRICULUM-TRAINED POLICIES



01 SELF-ARREST

STOP THE
SLIDE



02 FALL RECOVERY

GET BACK
UP



03 CONTROLLED RAPPEL

DESCEND WITH
CONTROL



04 TEAM LIFT

MOVE HEAVIER
LOADS TOGETHER

ONE POLICY. MANY FALLS.

MUJOCO SIMULATION / PPO / REWARD SHAPING

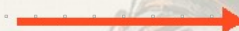
125 OBSERVATIONS

pose · speed · tool load



MUJOCO + PPO

100 Hz compact policy



14 ARM COMMANDS

plant and load the axe

PHYSICAL REWARD SHAPING IN MUJOCO

REWARDS REQUIRE RIGID PICK CONTACT, 22–42° BLADE ANGLE, AND REAL SNOW LOAD

CURRICULUM LEARNING PROGRESSION

EASY SLIDES → 5 m/s + CROSS-SLOPE + ADVERSARIAL FALLS

SUCCESS COUNTS ONLY WHEN THE AXE PHYSICALLY BITES.

ONE SHARED POLICY. EVERY ROBOT.

ISAAC SIM & ISAAC LAB / COOPERATIVE MAPPO



LOCAL STATE + TEAM TOKENS

98 local features + 7-D token / teammate



ISAAC SIM GPU ACTOR (MAPPO)

10 COMMANDS / G1 · FROZEN AGILE LEGS

TEAM REWARD SHAPING:

+ TRACK + LEVEL + LOAD BALANCE - RATE / FALLS

Same actor weights on every robot.

MULTI-AGENT CURRICULUM IN ISAAC SIM

LIFT FIRST → CARRY + TURN → HEAVIER LOADS (8→18 kg)

GPU-VECTORIZED SIMULATION TRAINS DECENTRALIZED COORDINATION

LIVEKIT IS THE EXPEDITION BUS.

STATE + VOICE

STATE

ONE STATE STREAM / G1



LIVEKIT ROOM



EVERY ACTOR STAYS IN SYNC

VOICE

"MOVE THE LOG"



CONFIRM



GATED START INTENT

VOICE SUPERVISES. LOCAL POLICIES CONTROL MOTION.

SIM-TO-REAL: OPEN CHALLENGES

THE C.L.I.M.B. ROADMAP FOR ALPINE ROBOTICS

C

CONTACT

SNOW & ICE

Slush, crust
fracture, and non-
linear friction
drops on steep
slopes

L

LIGHTING

OPTICS & SENSORS

Flat-light
whiteout
blindness, intense
snow glare, and
LiDAR backscatter

I

INTER-COMMS

OFFLINE MESH

Ad-hoc peer-to-
peer radio mesh
when cloud
connectivity is
unavailable

M

MECHANICAL

ROPES & TETHERS

Rope elasticity,
carabiner friction
variation, and
dynamic line
bounce

B

BALANCE

UNASSISTED TORQUE

Replacing sim-only
stabilizer
crutches with true
whole-body motor
control

CONTACT · LIGHTING · INTER-COMMS · MECHANICAL · BALANCE